

SITANSU DAS

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[LinkedIn](#) | [Github](#) | [Portfolio](#)

Professional Summary

Entry-level DevOps and cloud engineer (B.Sc. Computer Science) with three self-built, documented projects spanning container orchestration, serverless APIs, and system-health tooling. Skilled at taking code from commit to production through automated pipelines, codified infrastructure, and log-driven troubleshooting. Demonstrated autoscaling from 2 to 4 instances and automatic recovery of deleted pods in a 3-tier deployment. Targeting DevOps Engineer, Cloud Engineer, and Junior Site Reliability Engineer roles.

Technical Skills

Cloud: AWS (EC2, VPC, S3, IAM, Lambda, API Gateway)

Containers & Orchestration: Docker, Kubernetes

IaC & Configuration Management: Terraform, Ansible

CI/CD: GitHub Actions, Jenkins, GitLab CI

Monitoring & Logging: CloudWatch, Prometheus, Grafana

Scripting & APIs: Python, Bash, FastAPI, Boto3, YAML

OS, Data & Version Control: Linux, Nginx, PostgreSQL, DynamoDB, Git, GitHub, Jira

Networking & Support: TCP/IP, DNS, HTTP/HTTPS, Troubleshooting, Root Cause Analysis

Projects

Highly Scalable 3-Tier Application Deployment with Kubernetes & Terraform

Tech Stack: AWS (EC2, VPC), Terraform, Docker, Kubernetes, GitHub Actions, Nginx, PostgreSQL, Linux

- Architected a containerized 3-tier web application (frontend, API, PostgreSQL) on Kubernetes using 5 resource types (Deployments, Services, ConfigMaps, Secrets, Ingress), with Nginx routing external traffic.
- Provisioned the AWS network and compute layer (VPC, subnets, security groups, EC2 worker nodes) as Infrastructure as Code (IaC), making the environment reproducible without manual console work.
- Improved availability with 2 backend replicas, liveness and readiness probes, and a Horizontal Pod Autoscaler (HPA) that grew capacity to 4 instances under CPU load; confirmed that manually deleted pods were recreated automatically.
- Wired a GitHub Actions workflow to build images and roll out changes on every push, and documented architecture, provisioning steps, scaling tests, and recovery results in the repository.

Serverless Application Deployment & Automation using AWS Lambda

Tech Stack: AWS Lambda, API Gateway, DynamoDB, S3, Terraform, GitHub Actions, Python, CloudWatch

- Built a serverless REST API in Python with API Gateway routes and functions for create, read, update, and delete operations backed by DynamoDB, removing any need to manage application servers.
- Codified functions, permissions, routes, and tables as reusable templates, and scoped each IAM role to only the resources its function needs (least privilege).
- Automated validation, packaging, and infrastructure rollout through a Continuous Integration/Continuous Deployment (CI/CD) pipeline; kept settings in environment variables and static files in S3.
- Exercised the API with roughly 50-100 test requests, traced failures through CloudWatch Logs, and returned distinct HTTP status codes for success, invalid input, and application errors.

Internal DevOps Utilities API: System Health & AWS S3 Monitoring

Tech Stack: Python, FastAPI, Boto3, psutil, Uvicorn, Linux

- Designed a modular FastAPI service (router and service layers) with interactive Swagger and ReDoc docs, centralizing internal operations utilities for health monitoring and AWS visibility.
- Exposed a psutil health-check endpoint for live CPU, memory, and disk usage with configurable thresholds.
- Created a Boto3-powered S3 audit endpoint that inventories buckets and classifies each against a 90-day age threshold to support storage cleanup decisions.
- Established a log rotation strategy that deletes logs older than 90 days to control disk usage, with log backups scheduled through cron on Linux.

Certifications

AWS Cloud Practitioner Essentials (Course Completion Certificate), AWS Training & Certification

Sep 2026

Education

B.Sc. in Computer Science, Stewart Science Degree College, Utkal University | CGPA: 8.5

2022 - 2025

Higher Secondary Certificate (Science), SAIHSR, Cuttack, Odisha | 74.16%

2022